Tutorial Bluetooth ESP32 (Kontrol LED)

🎯 Tujuan

Siswa dapat:

- Mengirim perintah dari HP ke ESP32
- Mengontrol LED lewat Bluetooth

🧰 Alat & Bahan

- ESP32 (sudah ada Bluetooth bawaan)
- LED + resistor
- Kabel jumper
- HP (Android)
- Aplikasi Bluetooth Serial (contoh: Serial Bluetooth Terminal)

🔌 Wiring

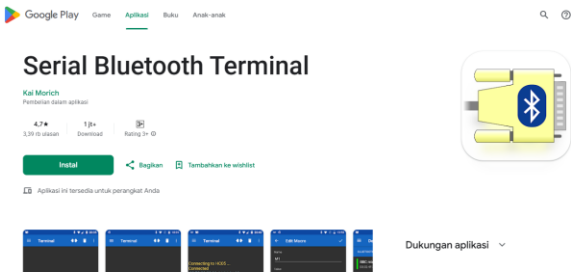
LED

- Kaki panjang → GPIO **19**
- Kaki pendek → resistor → GND

📱 Aplikasi HP

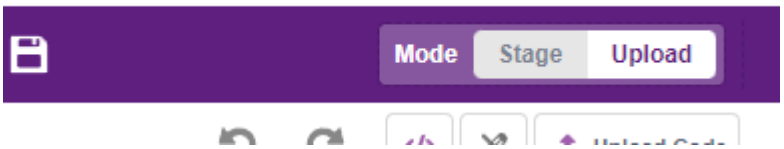
Install aplikasi:

- **Serial Bluetooth Terminal** (Android)



💻 Setup PictoBlox

1. Buka PictoBlox
2. Pilih **Board** → **ESP32**
3. Mode: **Upload Mode** (WAJIB untuk Bluetooth)



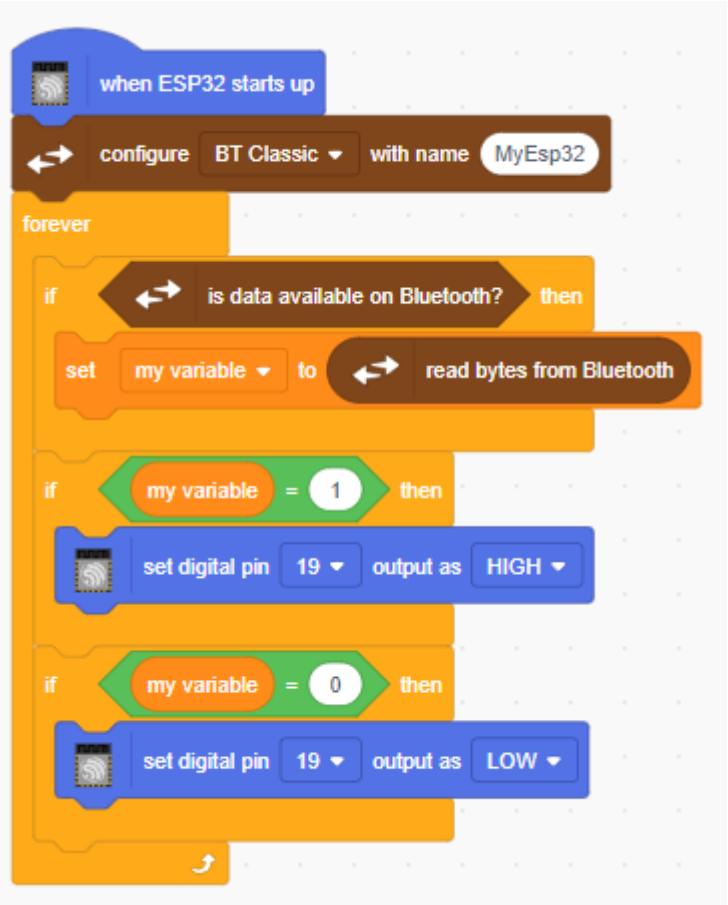
🧠 Konsep Program

HP kirim teks → ESP32 baca → LED dikontrol

Contoh:

- kirim 1 → LED nyala 💡
- kirim 0 → LED mati ❌

🧩 Program PictoBlox



Perbaiki kode program

```
8   BluetoothSerial esp32BT;
9
10  //Gloabl Variables are declared here
11  float my_variable;
12
13  void setup() {
14    //put your setup code here, to run once:
15    esp32BT.begin("MyEsp32");
16    pinMode(19, OUTPUT);
17
18
19  }
20
21  void loop() {
22    //put your main code here, to run repeatedly:
23
24
25    if(esp32BT.available()) {
26      my_variable = esp32BT.read();
27    }
28    if((my_variable == 1)) {
```

Klick edit mode dengan icon pensil

Kemudian ganti **float my_variable** menjadi **char my_variable**

Contoh setelah diperbaiki:

```
5
6
7   //MACROS are defined here
8   BluetoothSerial esp32BT;
9
10  //Gloabl Variables are declared here
11  char my_variable;
12
13  void setup() {
14    //put your setup code here, to run once:
15    esp32BT.begin("MyEsp32");
16    pinMode(19, OUTPUT);
17
18
```

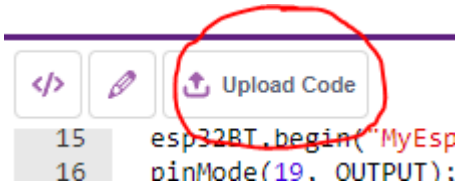
Selanjutanya Perbaiki kode dibawah ini **my_variable ==1** menjadi **my_variable ==’1’, my_variable ==0** menjadi **my_variable ==’0’**

```
void loop() {
  //put your main code here, to run


  if(esp32BT.available()) {
    my_variable = esp32BT.read();
  }
  if((my_variable == 1)) {
    digitalWrite(19, true);
  }
  if((my_variable == 0)) {
    digitalWrite(19, false);
  }
}
```

Sesudah diperbaiki

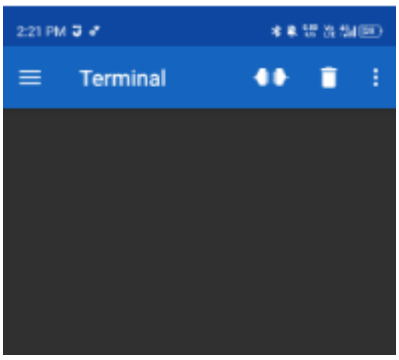
```
void loop() {
  //put your main code here, to ru


  if(esp32BT.available()) {
    my_variable = esp32BT.read();
  }
  if((my_variable == '1')) {
    digitalWrite(19, true);
  }
  if((my_variable == '0')) {
    digitalWrite(19, false);
  }
}
```

Kemudian uploa kode

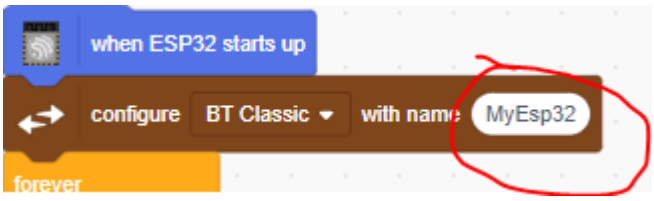


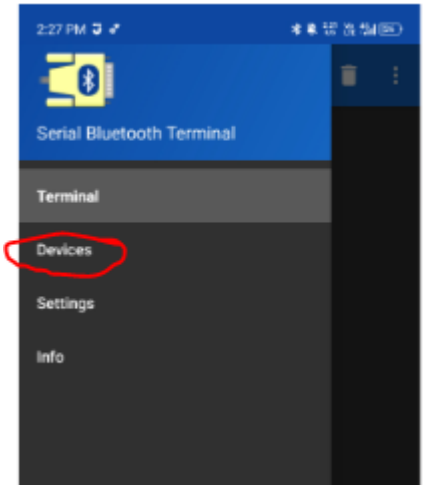
📲 Cara Menggunakan

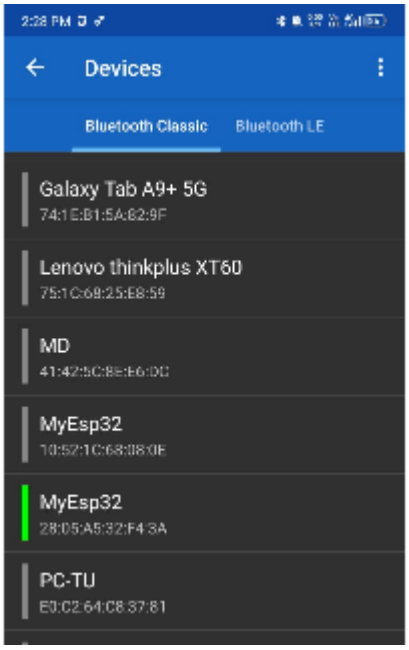
1. Buka aplikasi bluetottoh terminal
2. Nyalakan Bluetooth HP buka aplikasi serial bluetooth terminal di android



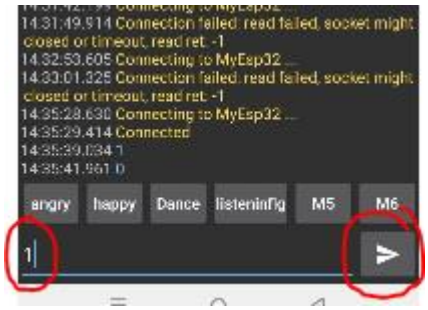
3. Hubungkan ke nama bluettooh (*disesuaikan dengan* nama yang kalian buat







4. Jika telah terhubung kirim data:



- 1 → LED nyala
- 2 → LED mati